

hello__t2.R

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```
library(tidyverse)
```

```
## -- Attaching packages ----- tidyverse 1.2.1 --
```

```
## v ggplot2 3.1.1      v purrr  0.3.2
## v tibble  2.1.1      v dplyr  0.8.0.1
## v tidyr   0.8.3      v stringr 1.4.0
## v readr   1.3.1      v forcats 0.4.0
```

```
## -- Conflicts ----- tidyverse_conflicts() --
```

```
## x dplyr::filter() masks stats::filter()
```

```
## x dplyr::lag()     masks stats::lag()
```

```
library(DBI)
```

```
## this is the magic database connection string
```

```
## read only access
```

```
## no problems with sharing this with others
```

```
con = RMySQL::dbConnect (
  drv      = RMySQL::MySQL(),
  dbname    = "pancan2018",
  host      = "pancan2018.cbe7mtbvw2d.us-east-1.rds.amazonaws.com",
  port      = 3306,
  username  = "reader",
  password  = "Readeruser19" )
```

```
## if you have a local sqlite database you can use this instead
```

```
if(FALSE){
  con = RSQLite::dbConnect(RSQLite::SQLite(), dbname = 'tcga.db', flags = RSQLite::SQLITE_RO)
}
```

```
#####
```

```
## load these convenient table bindings:
```

```
## you are connecting a relational database table to something that acts as a tibble/data.frame
##
```

```
## these first three are very important and capture
```

```
## almost the entire data in the db
```

```
tcgas = tbl(con, "tcgas") ## simple sample/probe/value/type view
```

```
tcgacats = tbl(con, "tcgacats") ## simple sample/probe/value/type view
```

```
clinpheno = tbl(con, "clinpheno") ## clinical/phenotype table
```

```
## below are less necessary, more convenience
```

```
## tcga = numerical genomic + clinical
```

```
tcga = tbl(con, "tcga") ## genomic + clinical columns
```

```
## sample list table
```

```
samples = tbl(con, "samples")
```

```
## Warning in .local(conn, statement, ...): Unsigned INTEGER in col 0 imported
## as numeric
```

```
## probe list table
probes = tbl(con, "probes")
```

```
## Warning in .local(conn, statement, ...): Unsigned INTEGER in col 0 imported
## as numeric
```

```
## vector of all probes
allprobes = probes %>% pull(probe)
## table of samples that were tested for mutation, will need this later
mutationsamples = tbl(con, 'mutationsamples') %>% pull(sample)
```

```
## example list of probes
myprobes = c('CCR8', 'IFNG', 'TP53.mut', 'PIK3CA.mut', 'PIK3CA.fmut')
```

```
## some properties of the clinical table
enframe(colnames(clinpheno))
```

```
## # A tibble: 46 x 2
##   name value
##   <int> <chr>
## 1     1 sample
## 2     2 sample_type_id
## 3     3 sample_type
## 4     4 cohort
## 5     5 Patient
## 6     6 tumtype
## 7     7 age_at_initial_pathologic_diagnosis
## 8     8 gender
## 9     9 race
## 10    10 ajcc_pathologic_tumor_stage
## # ... with 36 more rows
```

```
## summary breakdown of assigned subtype
clinpheno %>% select( Subtype_Selected ) %>%
  data.frame %>% table %>% as_tibble
```

```
## # A tibble: 126 x 2
##   . n
##   <chr> <int>
## 1 ACC.CIMP-high 20
## 2 ACC.CIMP-intermediate 27
## 3 ACC.CIMP-low 32
## 4 ACC.NA 13
## 5 AML.1 14
## 6 AML.2 21
## 7 AML.3 18
## 8 AML.4 37
## 9 AML.5 33
## 10 AML.6 35
## # ... with 116 more rows
```

```
## look at tcgas - the simple numeric table of sample/probe/type things
tcgas
```

```
## # Source:   table<tcgas> [?? x 4]
## # Database: mysql 5.7.25-log
## #   [reader@pancan2018.cbe7mtbvw2d.us-east-1.rds.amazonaws.com:/pancan2018]
##   sample      probe      value type
##   <chr>        <chr>      <dbl> <chr>
## 1 TCGA-OR-A5J1-01 100130426    0 rna
## 2 TCGA-OR-A5J2-01 100130426    0 rna
## 3 TCGA-OR-A5J3-01 100130426    0 rna
## 4 TCGA-OR-A5J5-01 100130426    0 rna
## 5 TCGA-OR-A5J6-01 100130426    0 rna
## 6 TCGA-OR-A5J7-01 100130426    0 rna
## 7 TCGA-OR-A5J8-01 100130426    0 rna
## 8 TCGA-OR-A5J9-01 100130426    0 rna
## 9 TCGA-OR-A5JA-01 100130426    0 rna
## 10 TCGA-OR-A5JB-01 100130426    0 rna
## # ... with more rows
```

```
## simple queries
tcgas %>% filter( probe == 'FOXP3' )
```

```
## # Source:   lazy query [?? x 4]
## # Database: mysql 5.7.25-log
## #   [reader@pancan2018.cbe7mtbvw2d.us-east-1.rds.amazonaws.com:/pancan2018]
##   sample      probe value type
##   <chr>        <chr> <dbl> <chr>
## 1 TCGA-OR-A5J1-01 FOXP3  6.66 rna
## 2 TCGA-OR-A5J2-01 FOXP3  4.04 rna
## 3 TCGA-OR-A5J3-01 FOXP3  3.22 rna
## 4 TCGA-OR-A5J5-01 FOXP3  4.37 rna
## 5 TCGA-OR-A5J6-01 FOXP3  2.86 rna
## 6 TCGA-OR-A5J7-01 FOXP3  2.82 rna
## 7 TCGA-OR-A5J8-01 FOXP3  5.45 rna
## 8 TCGA-OR-A5J9-01 FOXP3  2.77 rna
## 9 TCGA-OR-A5JA-01 FOXP3  4.14 rna
## 10 TCGA-OR-A5JB-01 FOXP3  5.53 rna
## # ... with more rows
```

```
tcgas %>% filter( probe == 'FOXP3.cnv' )
```

```
## # Source:   lazy query [?? x 4]
## # Database: mysql 5.7.25-log
## #   [reader@pancan2018.cbe7mtbvw2d.us-east-1.rds.amazonaws.com:/pancan2018]
##   sample      probe      value type
##   <chr>        <chr>      <dbl> <chr>
## 1 TCGA-A5-A0GI-01 FOXP3.cnv    0 cnv
## 2 TCGA-S9-A7J2-01 FOXP3.cnv    0 cnv
## 3 TCGA-VP-A878-01 FOXP3.cnv    0 cnv
## 4 TCGA-AR-A1AH-01 FOXP3.cnv    0 cnv
## 5 TCGA-EK-A2RE-01 FOXP3.cnv    0 cnv
```

```
## 6 TCGA-44-6778-01 FOXP3.cnv      0 cnv
## 7 TCGA-F4-6854-01 FOXP3.cnv      0 cnv
## 8 TCGA-74-6577-01 FOXP3.cnv      0 cnv
## 9 TCGA-AX-A2H7-01 FOXP3.cnv      0 cnv
## 10 TCGA-C8-A1HL-01 FOXP3.cnv      0 cnv
## # ... with more rows
```

```
tcgas %>% filter( probe == 'FOXP3.mut' )
```

```
## # Source:   lazy query [?? x 4]
## # Database: mysql 5.7.25-log
## #   [reader@pancan2018.cbe7mtbvw12d.us-east-1.rds.amazonaws.com:/pancan2018]
##   sample      probe      value type
##   <chr>        <chr>      <dbl> <chr>
## 1 TCGA-06-5416-01 FOXP3.mut      1 mut
## 2 TCGA-06-6699-01 FOXP3.mut      1 mut
## 3 TCGA-14-1823-01 FOXP3.mut      1 mut
## 4 TCGA-19-5956-01 FOXP3.mut      1 mut
## 5 TCGA-44-2655-01 FOXP3.mut      1 mut
## 6 TCGA-49-AAR0-01 FOXP3.mut      1 mut
## 7 TCGA-55-7910-01 FOXP3.mut      1 mut
## 8 TCGA-62-A46R-01 FOXP3.mut      1 mut
## 9 TCGA-A5-A2K5-01 FOXP3.mut      1 mut
## 10 TCGA-A6-5661-01 FOXP3.mut      1 mut
## # ... with more rows
```

```
tcgas %>% filter( probe %in% myprobes )
```

```
## # Source:   lazy query [?? x 4]
## # Database: mysql 5.7.25-log
## #   [reader@pancan2018.cbe7mtbvw12d.us-east-1.rds.amazonaws.com:/pancan2018]
##   sample      probe value type
##   <chr>        <chr> <dbl> <chr>
## 1 TCGA-OR-A5J1-01 CCR8      0 rna
## 2 TCGA-OR-A5J2-01 CCR8     2.33 rna
## 3 TCGA-OR-A5J3-01 CCR8      0 rna
## 4 TCGA-OR-A5J5-01 CCR8      0 rna
## 5 TCGA-OR-A5J6-01 CCR8      0 rna
## 6 TCGA-OR-A5J7-01 CCR8      0 rna
## 7 TCGA-OR-A5J8-01 CCR8     2.58 rna
## 8 TCGA-OR-A5J9-01 CCR8      0 rna
## 9 TCGA-OR-A5JA-01 CCR8      0 rna
## 10 TCGA-OR-A5JB-01 CCR8     1.22 rna
## # ... with more rows
```

```
## a simple spread() to get probes in columns
tcgas %>% filter( probe %in% myprobes ) %>%
  select( -type ) %>% collect %>%
  spread(probe, value)
```

```
## # A tibble: 11,224 x 5
##   sample      CCR8  IFNG PIK3CA.mut TP53.mut
```

```
##      <chr>          <dbl> <dbl>          <dbl>      <dbl>
## 1 TCGA-02-0003-01 NA      NA              NA          1
## 2 TCGA-02-0033-01 NA      NA              1          1
## 3 TCGA-02-0047-01 0.69  0              1          NA
## 4 TCGA-02-0055-01 0.64  1.08          NA          1
## 5 TCGA-02-2483-01 0.51  0.88          NA          1
## 6 TCGA-02-2485-01 0.56  1.54          1          1
## 7 TCGA-02-2486-01 0.69  1.15          NA          NA
## 8 TCGA-04-1348-01 2.85  4.55          NA          NA
## 9 TCGA-04-1357-01 4.18  5.43          NA          NA
## 10 TCGA-04-1362-01 6.05  0.15          NA          NA
## # ... with 11,214 more rows
```

```
## "tcga" get probe + clin results in tidy format
```

```
tcga %>% filter( probe == 'FOXP3' ) ## note: probe/value/type as last columns
```

```
## # Source:   lazy query [?? x 49]
## # Database: mysql 5.7.25-log
## #   [reader@pancan2018.cbe7mtbvwi2d.us-east-1.rds.amazonaws.com:/pancan2018]
##   sample sample_type_id sample_type cohort Patient tumtype
##   <chr>   <chr>          <chr>      <chr> <chr>   <chr>
## 1 TCGA-- 01              Primary Tu~ adren~ TCGA-0~ ACC
## 2 TCGA-- 01              Primary Tu~ adren~ TCGA-0~ ACC
## 3 TCGA-- 01              Primary Tu~ adren~ TCGA-0~ ACC
## 4 TCGA-- 01              Primary Tu~ adren~ TCGA-0~ ACC
## 5 TCGA-- 01              Primary Tu~ adren~ TCGA-0~ ACC
## 6 TCGA-- 01              Primary Tu~ adren~ TCGA-0~ ACC
## 7 TCGA-- 01              Primary Tu~ adren~ TCGA-0~ ACC
## 8 TCGA-- 01              Primary Tu~ adren~ TCGA-0~ ACC
## 9 TCGA-- 01              Primary Tu~ adren~ TCGA-0~ ACC
## 10 TCGA-- 01             Primary Tu~ adren~ TCGA-0~ ACC
## # ... with more rows, and 43 more variables:
## #   age_at_initial_pathologic_diagnosis <dbl>, gender <chr>, race <chr>,
## #   ajcc_pathologic_tumor_stage <chr>, clinical_stage <chr>,
## #   histological_type <chr>, histological_grade <chr>,
## #   initial_pathologic_dx_year <dbl>, menopause_status <chr>,
## #   birth_days_to <dbl>, vital_status <chr>, tumor_status <chr>,
## #   last_contact_days_to <dbl>, death_days_to <dbl>, cause_of_death <chr>,
## #   new_tumor_event_type <chr>, new_tumor_event_site <chr>,
## #   new_tumor_event_site_other <chr>, new_tumor_event_dx_days_to <dbl>,
## #   treatment_outcome_first_course <chr>, margin_status <chr>,
## #   residual_tumor <chr>, OS <dbl>, OS.time <dbl>, DSS <dbl>,
## #   DSS.time <dbl>, DFI <dbl>, DFI.time <dbl>, PFI <dbl>, PFI.time <dbl>,
## #   Redaction <chr>, Subtype_CNA <chr>, Subtype_DNAmeth <chr>,
## #   Subtype_Immune_Model_Based <chr>, Subtype_Integrative <chr>,
## #   Subtype_Selected <chr>, Subtype_mRNA <chr>, Subtype_miRNA <chr>,
## #   Subtype_other <chr>, Subtype_protein <chr>, probe <chr>, value <dbl>,
## #   type <chr>
```

```
## dimensions of results
```

```
tcga %>% filter( probe == 'FOXP3' ) %>% collect %>% dim
```

```
## [1] 11060    49
```

```

## a function to get you numeric genomics data
## from a list of probes
## and spread tidy data out into probe columns
genomedat = function(probes) {
  tcgas %>%
    filter( probe %in% probes ) %>%
    select( -type ) %>%
    collect %>% spread(probe, value)
}

genomedat( myprobes )

```

```

## # A tibble: 11,224 x 5
##   sample      CCR8  IFNG PIK3CA.mut TP53.mut
##   <chr>      <dbl> <dbl>      <dbl>      <dbl>
## 1 TCGA-02-0003-01 NA     NA          NA          1
## 2 TCGA-02-0033-01 NA     NA          1          1
## 3 TCGA-02-0047-01 0.69  0          1          NA
## 4 TCGA-02-0055-01 0.64  1.08       NA          1
## 5 TCGA-02-2483-01 0.51  0.88       NA          1
## 6 TCGA-02-2485-01 0.56  1.54       1          1
## 7 TCGA-02-2486-01 0.69  1.15       NA          NA
## 8 TCGA-04-1348-01 2.85  4.55       NA          NA
## 9 TCGA-04-1357-01 4.18  5.43       NA          NA
## 10 TCGA-04-1362-01 6.05  0.15       NA          NA
## # ... with 11,214 more rows

```

```

#####
## there's at least one catch for mutation
## the simple queries above aren't quite sufficient
## To save space, I *don't* store wt (0) mutations in the database.
## I *do* store a list of samples tested for mutations.
## one can add the wt calls back in
## btw:
## the lib.R file has the function gitr() that takes care
## of this and more for you.
## Here's an example set of functions though

## get all numeric probe data and impute wt in "mut"s
gdatnumeric = function(myprobes) {
  tcgas %>%
    filter( probe %in% myprobes ) %>% collect %>%
    select( -type ) %>%
    spread(probe, value) %>%
    mutate_at(
      vars( ends_with(".mut") ),
      funs( factor( case_when(
        is.na(.) & sample %in% mutationsamples ~ 'wt',
        !is.na(.) & . == 1 ~ 'mut',
        ) ) ) ) }

gdatnumeric(myprobes)

```

```

## Warning: funs() is soft deprecated as of dplyr 0.8.0

```

```
## please use list() instead
##
## # Before:
## funs(name = f(.))
##
## # After:
## list(name = ~f(.))
## This warning is displayed once per session.
```

```
## # A tibble: 11,224 x 5
##   sample      CCR8  IFNG PIK3CA.mut TP53.mut
##   <chr>      <dbl> <dbl> <fct>      <fct>
## 1 TCGA-02-0003-01 NA     NA     wt          mut
## 2 TCGA-02-0033-01 NA     NA     mut         mut
## 3 TCGA-02-0047-01 0.69  0     mut         wt
## 4 TCGA-02-0055-01 0.64  1.08 wt          mut
## 5 TCGA-02-2483-01 0.51  0.88 wt          mut
## 6 TCGA-02-2485-01 0.56  1.54 mut         mut
## 7 TCGA-02-2486-01 0.69  1.15 wt          wt
## 8 TCGA-04-1348-01 2.85  4.55 <NA>        <NA>
## 9 TCGA-04-1357-01 4.18  5.43 <NA>        <NA>
## 10 TCGA-04-1362-01 6.05  0.15 <NA>        <NA>
## # ... with 11,214 more rows
```

```
## get all categorical probe information and impute wt in "fmut"s
gdatcategorical = function(myprobes) {
  tcgacats %>%
    filter( probe %in% myprobes ) %>% collect %>%
    group_by( sample, probe, type ) %>%
    mutate( value = paste(value, collapse = ';') ) %>%
    ungroup %>%
    distinct( sample, probe, value, type ) %>%
    select( -type ) %>%
    spread( probe, value ) %>%
    mutate_at(
      vars( ends_with(".fmut") ),
      funs( factor( case_when(
        is.na(.) & sample %in% mutationsamples ~ 'wt',
        !is.na(.) ~ . ) ) ) )
    )
}
gdatcategorical(myprobes)
```

```
## # A tibble: 1,221 x 2
##   sample      PIK3CA.fmut
##   <chr>      <fct>
## 1 TCGA-02-0033-01 PIK3CA_p.E542K_Missense_Mutation
## 2 TCGA-02-0047-01 PIK3CA_p.N345K_Missense_Mutation
## 3 TCGA-02-2485-01 PIK3CA_p.E103G_Missense_Mutation
## 4 TCGA-05-4249-01 PIK3CA_p.E545K_Missense_Mutation
## 5 TCGA-05-4397-01 PIK3CA_p.E545K_Missense_Mutation
## 6 TCGA-06-0166-01 PIK3CA_p.L10_M16del_In_Frame_Del
## 7 TCGA-06-0171-01 PIK3CA_p.M1043V_Missense_Mutation
## 8 TCGA-06-0210-01 PIK3CA_p.E545K_Missense_Mutation
```

```
## 9 TCGA-06-0879-01 PIK3CA_p.E453_G460delinsDDF_In_Fram
## 10 TCGA-06-2570-01 PIK3CA_p.E542V_Missense_Mutation
## # ... with 1,211 more rows
```

*## the two functions above can be strung together with joins
you can add the clinpheno clinical table too...*

```
num_and_cat = function( myprobes ) {
  gdatnumeric(myprobes) %>%
    full_join( gdatcategorical(myprobes) )
}

num_and_cat(myprobes)
```

```
## Joining, by = "sample"
```

```
## # A tibble: 11,224 x 6
##   sample      CCR8  IFNG PIK3CA.mut TP53.mut PIK3CA.fmut
##   <chr>      <dbl> <dbl> <fct>      <fct>      <fct>
## 1 TCGA-02-0003~ NA     NA    wt          mut         <NA>
## 2 TCGA-02-0033~ NA     NA    mut         mut         PIK3CA_p.E542K_Missense_M~
## 3 TCGA-02-0047~ 0.69  0     mut         wt          PIK3CA_p.N345K_Missense_M~
## 4 TCGA-02-0055~ 0.64  1.08 wt          mut         <NA>
## 5 TCGA-02-2483~ 0.51  0.88 wt          mut         <NA>
## 6 TCGA-02-2485~ 0.56  1.54 mut         mut         PIK3CA_p.E103G_Missense_M~
## 7 TCGA-02-2486~ 0.69  1.15 wt          wt          <NA>
## 8 TCGA-04-1348~ 2.85  4.55 <NA>        <NA>        <NA>
## 9 TCGA-04-1357~ 4.18  5.43 <NA>        <NA>        <NA>
## 10 TCGA-04-1362~ 6.05  0.15 <NA>        <NA>        <NA>
## # ... with 11,214 more rows
```

get it all, at least for the probes you ask for

```
alldat = function( myprobes ) {
  gdatnumeric(myprobes) %>%
    full_join( gdatcategorical(myprobes) ) %>%
    left_join( clinpheno %>% collect )
}

alldat(myprobes)
```

```
## Joining, by = "sample"
```

```
## Joining, by = "sample"
```

```
## # A tibble: 11,224 x 51
##   sample CCR8  IFNG PIK3CA.mut TP53.mut PIK3CA.fmut sample_type_id
##   <chr> <dbl> <dbl> <fct>      <fct>      <fct>      <chr>
## 1 TCGA~~ NA     NA    wt          mut         <NA>        01
## 2 TCGA~~ NA     NA    mut         mut         PIK3CA_p.E~ 01
## 3 TCGA~~ 0.69  0     mut         wt          PIK3CA_p.N~ 01
## 4 TCGA~~ 0.64  1.08 wt          mut         <NA>        01
```



```
## 5 TCGA-- 0.51 0.88 wt      mut      <NA>      01
## 6 TCGA-- 0.56 1.54 mut      mut      PIK3CA_p.E~ 01
## 7 TCGA-- 0.69 1.15 wt      wt      <NA>      01
## 8 TCGA-- 2.85 4.55 <NA>    <NA>    <NA>      01
## 9 TCGA-- 4.18 5.43 <NA>    <NA>    <NA>      01
## 10 TCGA-- 6.05 0.15 <NA>    <NA>    <NA>      01
## # ... with 11,214 more rows, and 44 more variables: sample_type <chr>,
## #   cohort <chr>, Patient <chr>, tumtype <chr>,
## #   age_at_initial_pathologic_diagnosis <dbl>, gender <chr>, race <chr>,
## #   ajcc_pathologic_tumor_stage <chr>, clinical_stage <chr>,
## #   histological_type <chr>, histological_grade <chr>,
## #   initial_pathologic_dx_year <dbl>, menopause_status <chr>,
## #   birth_days_to <dbl>, vital_status <chr>, tumor_status <chr>,
## #   last_contact_days_to <dbl>, death_days_to <dbl>, cause_of_death <chr>,
## #   new_tumor_event_type <chr>, new_tumor_event_site <chr>,
## #   new_tumor_event_site_other <chr>, new_tumor_event_dx_days_to <dbl>,
## #   treatment_outcome_first_course <chr>, margin_status <chr>,
## #   residual_tumor <chr>, OS <dbl>, OS.time <dbl>, DSS <dbl>,
## #   DSS.time <dbl>, DFI <dbl>, DFI.time <dbl>, PFI <dbl>, PFI.time <dbl>,
## #   Redaction <chr>, Subtype_CNA <chr>, Subtype_DNAmeth <chr>,
## #   Subtype_Immune_Model_Based <chr>, Subtype_Integrative <chr>,
## #   Subtype_Selected <chr>, Subtype_mRNA <chr>, Subtype_miRNA <chr>,
## #   Subtype_other <chr>, Subtype_protein <chr>
```

```
#####
## in addition to dplyr, you can use sql just as easily
```

```
library(sqldf)
```

```
## Loading required package: gsubfn
```

```
## Loading required package: proto
```

```
## Could not load tcltk. Will use slower R code instead.
```

```
## Loading required package: RSQLite
```

```
options(sqldf.connection = con)
```

```
sqldf('select * from tcgas limit 1')
```

```
##           sample      probe value type
## 1 TCGA-OR-A5J1-01 100130426      0  rna
```

```
sqldf('select * from tcgas where probe = "FOXP3" limit 1')
```

```
##           sample probe value type
## 1 TCGA-OR-A5J1-01 FOXP3  6.66  rna
```

```
sqldf('select * from tcgacats limit 5')
```

```
##           sample      probe           value
## 1 TCGA-OR-A5J1-01 Subtype_mRNA steroid-phenotype-high+proliferatio
## 2 TCGA-OR-A5J2-01 Subtype_mRNA steroid-phenotype-high+proliferatio
## 3 TCGA-OR-A5J3-01 Subtype_mRNA           steroid-phenotype-high
## 4 TCGA-OR-A5J4-01 Subtype_mRNA
## 5 TCGA-OR-A5J5-01 Subtype_mRNA           steroid-phenotype-high
##           type
## 1 molec_subtype
## 2 molec_subtype
## 3 molec_subtype
## 4 molec_subtype
## 5 molec_subtype
```

```
sqldf('select * from tcgacats where type = "fmut" limit 5')
```

```
##           sample      probe           value type
## 1 TCGA-02-0003-01 TACC2.fmut TACC2_p.T38M_Missense_Mutation fmut
## 2 TCGA-02-0003-01 JAKMIP3.fmut JAKMIP3_p.D723D_Silent fmut
## 3 TCGA-02-0003-01 PANX3.fmut PANX3_p.R296Q_Missense_Mutation fmut
## 4 TCGA-02-0003-01 SPI1.fmut SPI1_p.P127T_Missense_Mutation fmut
## 5 TCGA-02-0003-01 NAALAD2.fmut NAALAD2_p.R65C_Missense_Mutation fmut
```

```
sqldf('
select value from tcgacats
where type = "molec_subtype"
and probe = "Subtype_Selected") %>%
  table %>% as_tibble
```

```
## # A tibble: 99 x 2
##       .           n
##   <chr>       <int>
## 1 ACC.CIMP-high      20
## 2 ACC.CIMP-intermediate 27
## 3 ACC.CIMP-low       32
## 4 ACC.NA             12
## 5 AML.1              14
## 6 AML.2              21
## 7 AML.3              18
## 8 AML.4              37
## 9 AML.5              33
## 10 AML.6             35
## # ... with 89 more rows
```

```
#####
## do you want an entire set of genomics data?
## you'll need to be patient and have RAM
## a good network to east coast amazon also helps
## data.table library is recommended for spread/cast
library(data.table)
```

```
## data.table 1.12.2 using 1 threads (see ?getDTthreads). Latest news: r-datatable.com
```

```
## *****
```

```
## This installation of data.table has not detected OpenMP support. It should still work but in single-
```

```
## *****
```

```
##
```

```
## Attaching package: 'data.table'
```

```
## The following objects are masked from 'package:dplyr':
```

```
##
```

```
##      between, first, last
```

```
## The following object is masked from 'package:purrr':
```

```
##
```

```
##      transpose
```

```
Sys.time()
```

```
## [1] "2019-05-20 15:56:43 PDT"
```

```
n = 100000 ## testing only, set to Inf for query to get all rna
```

```
d = tcgas %>%
```

```
  filter( type == "rna" ) %>%
```

```
    head(n) %>%
```

```
      select(sample, probe, value) %>%
```

```
        collect
```

```
## it is unfortunate you cannot cast nor spread in sql
```

```
d = dcast( setDT(d), sample ~ probe, value.var = "value")
```

```
Sys.time()
```

```
## [1] "2019-05-20 15:56:44 PDT"
```

```
dim(d)
```

```
## [1] 11069    11
```

```
d[1:3,1:3]
```

```
##           sample 100130426 100133144
```

```
## 1: TCGA-02-0047-01          0      2.03
```

```
## 2: TCGA-02-0055-01          0      0.00
```

```
## 3: TCGA-02-2483-01          0      1.98
```

```
## for mutation, if we pull the full .mut data
```

```
## there's an easier way to impute zeroes
```

```
## this doesn't take that long
```

```
e = tcgas %>%
```

```
  filter( type == "mut" ) %>%
```

```
    select(-type) %>%
```

```
      collect %>%
```

```
        spread(probe, value, fill = 0)
```

```
Sys.time()
```

```
## [1] "2019-05-20 15:57:31 PDT"
```

```
dim(e)
```

```
## [1] 9093 19622
```

```
e[1:5,1:5]
```

```
## # A tibble: 5 x 5
##   sample      A1BG.mut A1CF.mut A2M.mut A2ML1.mut
##   <chr>      <dbl>    <dbl>    <dbl>    <dbl>
## 1 TCGA-02-0003-01      0        0        0        0
## 2 TCGA-02-0033-01      0        0        0        0
## 3 TCGA-02-0047-01      0        0        0        0
## 4 TCGA-02-0055-01      0        0        0        0
## 5 TCGA-02-2470-01      0        0        0        0
```

```
#####
## simple plots
library(ggplot2)

ggplot( aes(x = FOXP3, y = ABCA1, color = OS, size = OS.time),
  data = alldat( c("FOXP3", "ABCA1") ) ) +
  geom_point( alpha = 0.1 ) +
  geom_smooth(method = 'lm') +
  facet_wrap( ~ tumtype )
```

```
## Joining, by = "sample"
```

```
## Joining, by = "sample"
```

```
## Warning: Removed 117 rows containing missing values (geom_point).
```

